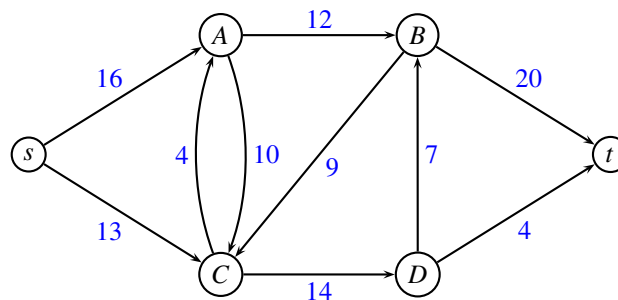


Solution, exam 19 December, 2018.

Course title: Network Optimization

Course no: 42115

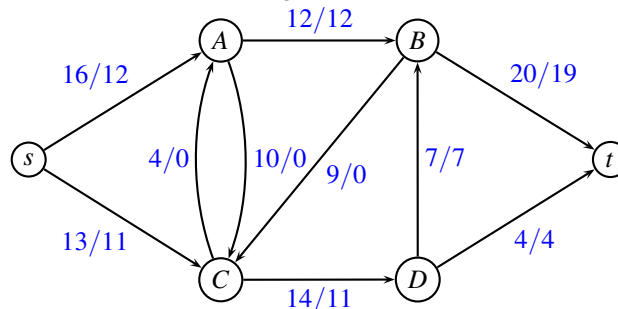
Maximum Flow problem



Answer 1 We solve the maximum flow problem using an augmenting path algorithm.

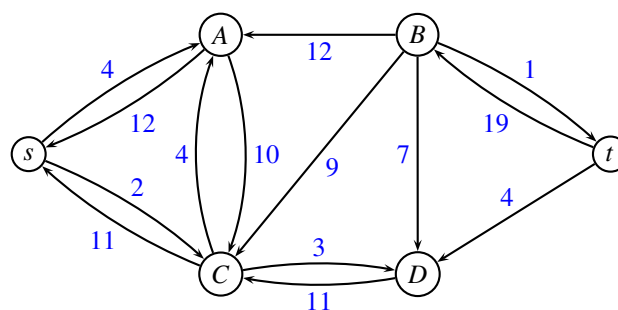
- path $sABt$ capacity 12
- path $sCDBt$ capacity 7
- path $sCDt$ capacity 4

No more augmenting paths can be found. This gives the solution:



with flow $12 + 11 = 23$. ■

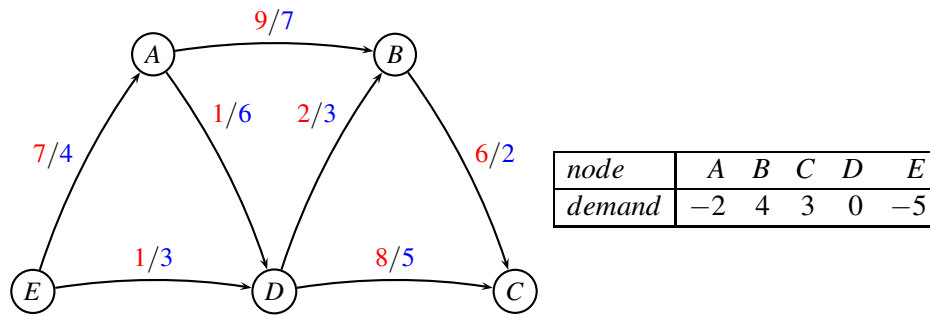
Answer 2 The residual graph is:



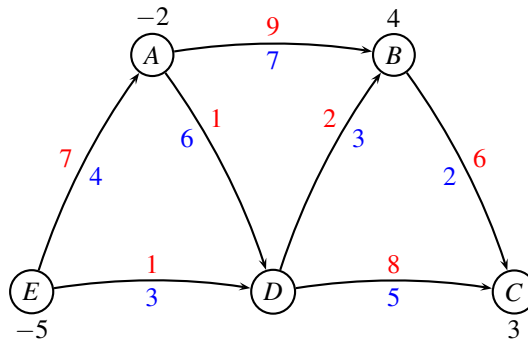
A minimum cut is $S = \{s, A, C, D\}$ and $T = \{B, t\}$. The capacity of the cut is $12 + 7 + 4 = 23$. ■

Empty container repositioning

MEDITLINE: Route net of a Mediterranean liner shipping company



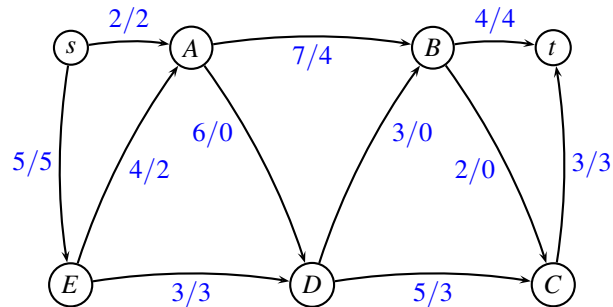
Answer 3 Since all containers are the same, we formulate the problem as a min cost flow problem, and solve it using the augmenting cycle algorithm or network simplex algorithm. The demands are set to $b_i := d_i$. The network becomes:



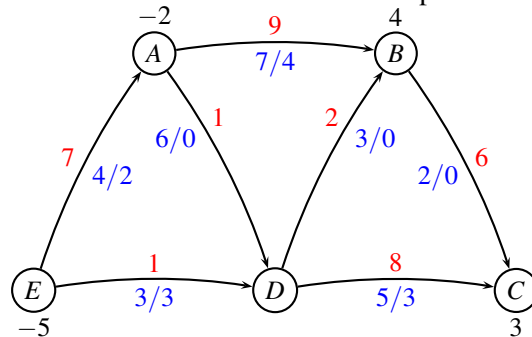
■

Answer 4 In order to find an initial solution we solve a maximum flow problem. Augmenting paths are: $sEDCt$ with capacity 3. $sABt$ with capacity 2. $sEABt$ with capacity 2. This is illustrated in the

following:

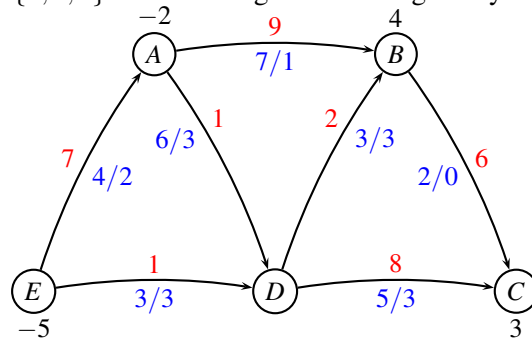


This leads to the following initial solution to the min cost flow problem:



Cost of the solution is $z = 77$.

We use the *augmenting cycle* algorithm. An augmenting cycle is $ADBA$ with cost $1 + 2 - 9 = -6$. Capacity of the cycle is $\min\{6, 3, 4\} = 3$. Sending 3 units along the cycle gives



No more negative cycles can be found, so optimal solution found. Cost of the solution is $z = 59$.

Note: One can also use the *Network Simplex* algorithm. In this case an initial solution should be found as described above, and then the algorithm is run. ■

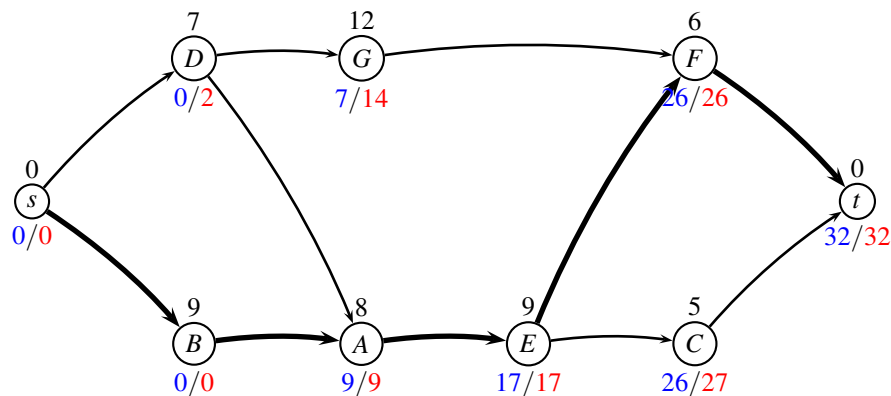
Critical path problem

Activity i	Description	Predecessors	Duration d_i
A	Start oven at 250 degrees	D, B	8
B	Weigh rye flour and wheat flour		9
D	Dissolve yeast in the water		7
C	Prepare a barrel for storage	E	5
E	Cut the dough into squares	A	9
F	Bake the biscuits	G, E	6
G	Mix ingredients and add salt	D	12

Answer 5 A possible ordering of the activities is: *sDBGAEF**Ct*. This is also illustrated in the graph below. ■

Answer 6 Now we solve the problem

- *Forward procedure:* Starting at time zero, calculate the earliest time U_j each activity j can be started $U_j = \max_{(i,j) \in E} \{U_i + d_i\}$. (marked with blue)
- *Makespan:* $T = U_t = 32$.
- *Backward procedure:* Starting at time equal to the makespan, calculate the latest time V_j each activity j can be started so that this makespan is realized. $V_i = \min_{(i,j) \in E} \{V_j - d_i\}$ (marked with red)



Answer 7 An activity j is on the critical path if $U_j = V_j$. The critical path is marked with fat edges. There is only one critical path. ■

I found the following recepy for making ship's biscuits. It is a recepy from the 18th century. Unfortunately it is in Danish, so you must use *Google translate* to get it in English.

Beskøjter

650 gram rugmel
350 gram hvedemel
50 gram gær
15 gram salt
1 liter vand

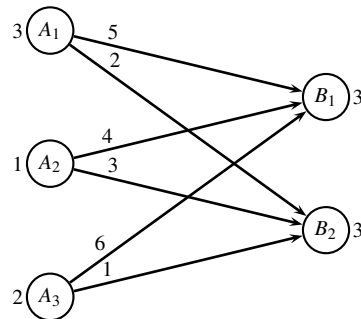
Gæren opløses i vandet, det øvrige blandes i og æltes godt. Om nødvendigt tilsættes lidt mere hvedemel for at gøre dejen egnet til udrulning.

Dejen stilles til hævnning $\frac{1}{2}$ -1 time, udrulles til plader på ca. 1 centimeters tykkelse, skæres i firkanter og sættes på plader med bagepapir. Beskøjterne efterhæves og prikkes med en gaffel.

Bages for 250 grader i et kvarter og tørres derefter for svag varme og åben luge i 1-3 timer, alt efter om de skal have et blødt indre eller være helt hårde.

Opskriften stammer fra 1800-tallet og findes på Folkklivsarkivet i Lund.

Fixed charge transportation problem



Answer 8 We have the following possible patterns

	A_1				A_2		A_3		
B_1	3	2	1	0	1	0	2	1	0
B_2	0	1	2	3	0	1	0	1	2
transp. cost	15	12	9	6	4	3	12	7	2
fixed charges	4	8	8	4	4	4	4	8	4
sum of costs	19	20	17	10	8	7	16	15	6

This leads to the Dantzig-Wolfe decomposed model:

```

minimize
19p11+20p12+17p13+10p14+ 8p21+7p22+16p31+15p32+ 6p33
subject to
3p11+ 2p12+ 1p13+ 0p14+ 1p21+0p22+ 2p31+ 1p32+ 0p33 = 3    (u1)
0p11+ 1p12+ 2p13+ 3p14+ 0p21+1p22+ 0p31+ 1p32+ 2p33 = 3    (u2)

p11+  p12+  p13+  p14                                     = 1    (v1)
                                p21 +p22                     = 1    (v2)
                                p31+  p32+  p33               = 1    (v3)

binary p11 p12 p13 p14 p21 p22 p31 p32 p33
end

```

■

Answer 9 The initial solution corresponds to the restricted master problem

```

minimize
    20p12                                + 7p22                                +15p32
subject to
    2p12                                + 0p22                                + 1p32                                = 3    (u1)
    1p12                                + 1p22                                + 1p32                                = 3    (u2)

    p12                                = 1    (v1)
                                p22                                = 1    (v2)
                                p32                                = 1    (v3)

end

```

We are told that the dual variables for the restricted master problem are $u_1 = 0, u_2 = 0, v_1 = 20, v_2 = 7, v_3 = 15$.

Let a_{ip} be the coefficients in the two first lines of the pattern formulation, and b_{ip} be the coefficients in the last three lines. Reduced cost for pattern p is then calculated as

$$\bar{c}_p = c_p - \sum_i a_{pi}u_i - \sum_i b_{pi}v_i$$

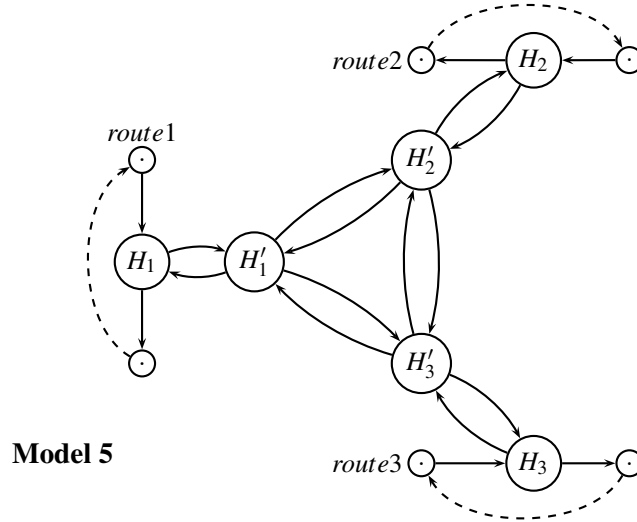
This gives the following reduced costs (we do not need to calculate reduced costs of patterns that are present in the restricted master problem)

pattern p	c_p	$\sum_i a_{pi}u_i$	$\sum_i b_{pi}v_i$	$\bar{c}_p$
p_{11}	19	0	20	-1
p_{13}	17	0	20	-3
p_{14}	10	0	20	-10
p_{21}	8	0	7	1
p_{31}	16	0	15	1
p_{33}	6	0	15	-9

Pattern p_{14} has the most negative reduced cost, so it should be added to the restricted master problem.

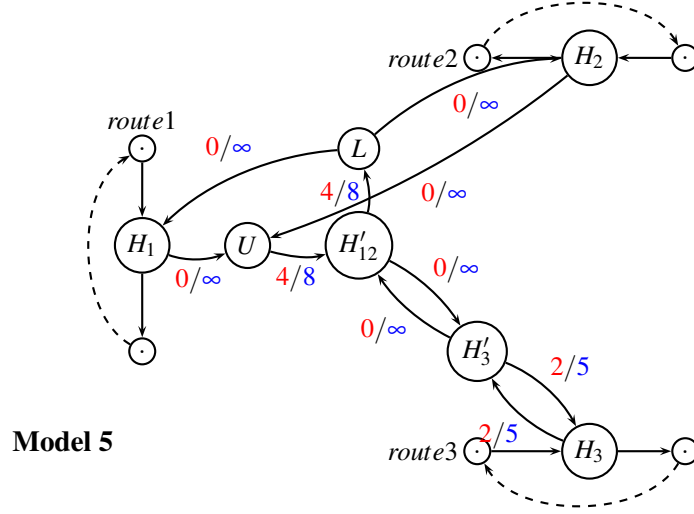
■

Modeling a hub port



We are considering Model 5 depicted above. Two vessels arrive the same day (route 1) and (route 2). Therefore, they have to share the crane capacity for loading and unloading. This means that the number of containers unloaded from vessels at route1 *and* route2 together should be below $u = 8$. Similarly, the number of containers loaded to the two routes together should be below $u = 8$.

Answer 10 Since the two vessels arrive the same day, we can merge nodes H'_1 and H'_2 into one node H'_{12} . Moreover we introduce two auxiliary nodes, L and U for loading and unloading. The capacity between U and H'_{12} , and between H'_{12} and L is set to $u = 8$. The capacity between H_1 and U, L , and between U, L and H_2 is set to $u = \infty$. The costs should be assigned in a similar way.



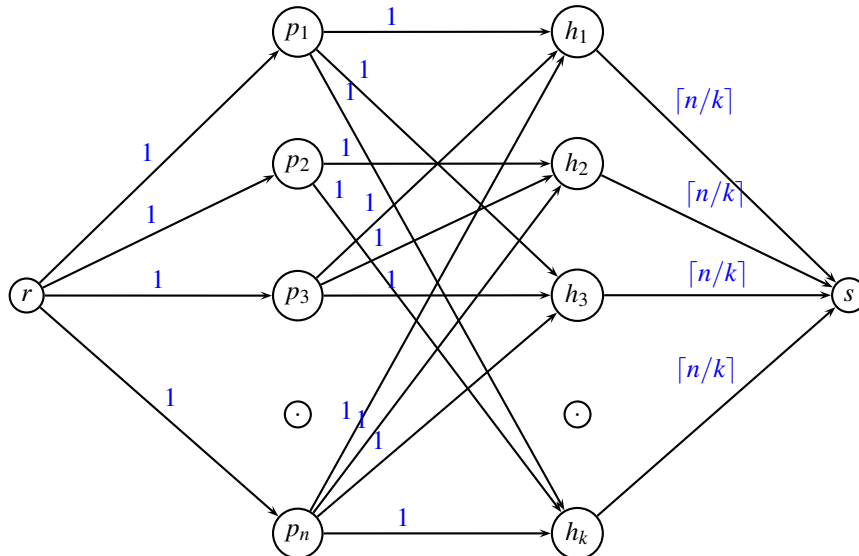
Model 5

Note: It is not sufficient to merge nodes H'_1 and H'_2 into one node H'_{12} , without introducing nodes U and L . Otherwise, it is not possible to model the shared capacity (or you will be able to transship an unlimited number of containers from route1 to route2). ■

Flooding

Answer 11 We have n people, and k hospitals. Each hospital may take at most $\lceil n/k \rceil$ people. We model the problem as a *maximum flow problem* on a bi-partite graph (P, H) . Let the set of nodes P correspond to the people, and the set of nodes H correspond to hospitals. For every patient p_i and hospital h_j we have an edge with capacity 1, if the patient can be brought to the hospital within a half-hour's drive. If the patient cannot be brought to the hospital within a half-hour's drive, the edge is omitted.

Introduce a root r and a sink s . The root r is connected to all patients p_i with capacity 1. All hospitals h_j are connected to sink s with capacity $\lceil n/k \rceil$.

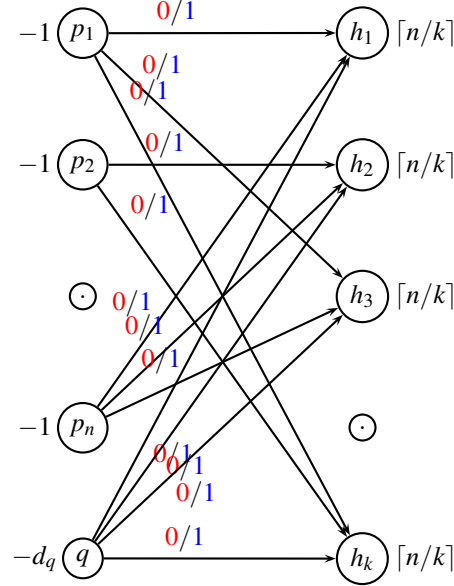


If a flow can be found covering all patients (i.e. flow of n) then it is possible to assign all patients to hospitals.

Alternative solution: We model the problem as a *min cost flow problem* on a bi-partite graph (P, H) . To have a balanced problem, we add a dummy patient node q to adjust for non-integer demands.

For every patient p_i and hospital h_j we have an edge with capacity 1 and cost 0, if the patient can be brought to the hospital within a half-hour's drive. If the patient cannot be brought to the hospital within a half-hour's drive, the edge is omitted. The dummy patient node q is connected to all hospitals h_j with an edge having capacity 1 and cost 0.

The demand of patient p_i is $d_i = -1$, while the demand for hospital h_j is $d_j = \lceil n/k \rceil$. The demand for dummy patient node q is $-d_q = -(k \cdot \lceil n/k \rceil - n)$.

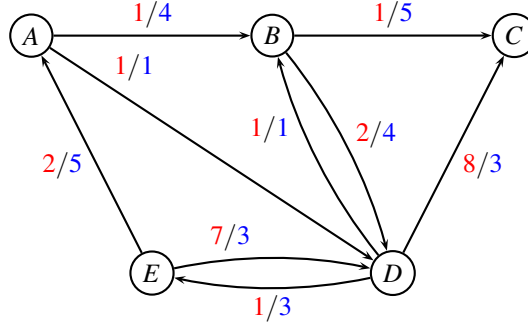


The problem is solved as a min cost flow problem which is solvable in polynomial time. If a feasible solution can be found, then it is possible to find an allocation of people to hospitals.

Note: It is not sufficient to have demand n/k for the hospitals, since if this is a fractional number, some patients will need to be split into fractions. ■

Unsplittable multi-commodity flow problem

ASIANET (Route net of a liner shipping company)



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij} . Cities are $A = \text{Manilla}$, $B = \text{Shanghai}$, $C = \text{HongKong}$, $D = \text{Xiamen}$, $E = \text{Singapore}$.

We have the following *restricted master problem*:

$$\begin{array}{llllllll}
 \min & 10x_1 & + & 16x_2 & + & 5x_3 & + & 15x_4 \\
 \text{s.t.} & 2x_1 & & & & & & \leq 4 & (y_{AB}) \\
 & & & & x_3 & & & \leq 1 & (y_{AD}) \\
 & & & & x_3 & & & \leq 5 & (y_{BC}) \\
 & 2x_1 & & & & & & \leq 4 & (y_{BD}) \\
 & & & & x_3 & & & \leq 1 & (y_{DB}) \\
 & & 2x_2 & & & + & x_4 & \leq 3 & (y_{DC}) \\
 & & & & & & & \leq 3 & (y_{DE}) \\
 & & & & x_3 & & & \leq 5 & (y_{EA}) \\
 & 2x_1 & & & & + & x_4 & \leq 3 & (y_{ED}) \\
 & x_1 & & & & & & \geq 1 & (\bar{y}_1) \\
 & & x_2 & & & & & \geq 1 & (\bar{y}_2) \\
 & & & & x_3 & + & x_4 & \geq 1 & (\bar{y}_3) \\
 & & & & & & & 0 \leq x_1, x_2, x_3, x_4 \leq 1
 \end{array} \tag{1}$$

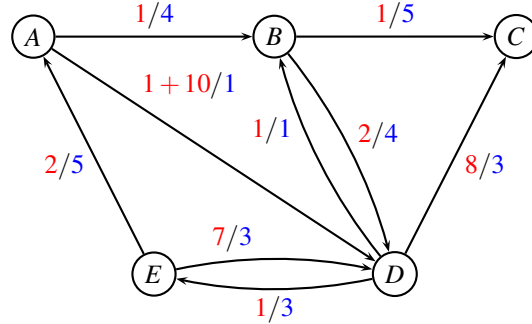
The dual solution to the model is: $y_{AD} = -10$, $\bar{y}_1 = 10$, $\bar{y}_2 = 16$, $\bar{y}_3 = 15$, while all other dual variables are zero.

Answer 12 The reduced cost of a route r is calculated as

$$\bar{c}_r^k = \sum_{(i,j) \in r} d_k(c_{ij} - y_{ij}) - \bar{y}_k$$

We can solve the pricing problem as a shortest path problem, by setting the cost of edge (i, j) to

$(c_{ij} - y_{ij})$. In our case the graph becomes:



Edges with capacity less than d_k are removed before solving the shortest path problem for commodity k .

Notice that $y_{ij} \leq 0$ since the restricted master problem is a minimization problem, and the constraints are “smaller-equal” constraints. This means that all edge costs will be non-negative, and we can therefore safely use Dijkstra’s algorithm. Once the shortest path problem is solved with solution value z , the reduced cost can be calculated as $\bar{c} = d_k \cdot z - \bar{y}_k$. ■

Answer 13 We solve the shortest path problem using Dijkstra’s algorithm

$k = 1$: We remove edges (AD) , (DB) , and find shortest path EABD (route 1) with cost $z = 5$. Reduced cost is $d_1 \cdot z - \bar{y}_1 = 2 \cdot 5 - 10 = 0$.

$k = 2$: We remove edges (AD) , (DB) , and find shortest path DEABC (route 4) with cost $z = 5$. Reduced cost is $d_2 \cdot z - \bar{y}_2 = 2 \cdot 5 - 16 = -6$.

$k = 3$: We do not remove any edges. Shortest path EABC (route 5) with cost $z = 4$. Reduced cost is $d_3 \cdot z - \bar{y}_3 = 1 \cdot 4 - 15 = -11$.

The most negative reduced cost is obtained with path EABC (route 5).

The resulting model becomes

$$\begin{array}{llllllllllllllll}
 \min & 10x_1 & + & 16x_2 & + & 5x_3 & + & 15x_4 & + & 4x_5 & & & & & & & & & \\
 \text{s.t.} & 2x_1 & & & & & & & & + & x_5 & \leq & 4 & & & & & & (y_{AB}) \\
 & & & & & x_3 & & & & & & \leq & 1 & & & & & & (y_{AD}) \\
 & & & & x_3 & & & + & x_5 & \leq & 5 & & & & & & & & (y_{BC}) \\
 & 2x_1 & & & & & & & & & & \leq & 4 & & & & & & (y_{BD}) \\
 & & & & x_3 & & & & & & & \leq & 1 & & & & & & (y_{DB}) \\
 & & 2x_2 & & & + & x_4 & & & & & \leq & 3 & & & & & & (y_{DC}) \\
 & & & & & & & & & & & \leq & 3 & & & & & & (y_{DE}) \\
 & & & & x_3 & & & + & x_5 & \leq & 5 & & & & & & & & (y_{EA}) \\
 & 2x_1 & & & & + & x_4 & & & \leq & 3 & & & & & & & & (y_{ED}) \\
 & x_1 & & & & & & & & & & \geq & 1 & & & & & & (\bar{y}_1) \\
 & & x_2 & & & & & & & & & \geq & 1 & & & & & & (\bar{y}_2) \\
 & & & x_3 & + & x_4 & + & x_5 & \geq & 1 & & & & & & & & & (\bar{y}_3) \\
 & 0 \leq x_1, x_2, x_3, x_4, x_5 \leq 1 & & & & & & & & & & & & & & & & &
 \end{array} \tag{2}$$

■